

Raagav Ramakrishnan

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ML Systems Engineering • Agentic AI • HCI

Education

International Institute of Information Technology, Hyderabad (IIIT-H) 2022 – 2026
B.Tech Honours in Electronics and Communication

The Millennium School, Dubai June 2022
High School Diploma CBSE — Grade 10: 93%, Grade 12: 95%

Work Experience

Undergraduate Researcher Jan. 2025 – Present
Data Sciences and Analytics Center (DSAC), IIIT Hyderabad *Advisor: Prof. Nimmi Rangaswamy*

- Investigating how young adults in urban India assemble hybrid fitness-tracking ecologies across wearables, apps, gym routines, social media, trainer feedback, and embodied self-knowledge.
- Conducted field visits and semi-structured interviews with gym-goers to study motivation, identity formation, progress validation, and well-being grounded against Indian socio-material contexts.
- Used affinity mapping, thematic clustering, and HCI literature synthesis to analyze how users negotiate between quantified self-tracking, social influence, routine disruption, and intuitive/embodied judgement.
- Contributed to framing design implications for adaptive personal informatics systems that support approximate logging, routine breaks, private accountability, and culturally situated fitness practices.
- **Publication:** Singh, S., **Ramakrishnan, R.**, Mahipal, C., Rangaswamy, N. (April 2026). *Everyday HCI of Adaptive Fitness: The Bricolage of Self-Tracking in Urban India*. In *CHI EA '26: Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*. ACM.

ML Research Engineer Dec. 2024 – Present
Cloud Ambassadors (Startup within the Google Ecosystem)

- Implemented continuous training and automated inference workflows using Vertex AI, Pub/Sub triggers, and Cloud Build CI/CD, with emphasis on reproducibility and rollout readiness.
- Developed a Dynamic Pricing Engine using XGBoost, FastAPI, Docker, BigQuery, Kubeflow, and GCP services for training, inference, and model versioning.
- Built a glioma detection and segmentation pipeline for 3D-MRI workflows on GCP, with native PACS to connect model inference with clinical deployment considerations.
- Fine-tuned ProtBERT and ESM-1b models for mutation effect prediction; integrated Graph Neural Networks (Graphormer/MPNN) using protein contact maps to model structural dependencies.

Projects

Neuroanatomy SME Agent | *HuggingFace, PyTorch, LangChain* *Nov. 2025*

- Implemented a full embedding and retrieval pipeline — batch ingestion, multi-granularity chunk-graph construction, BioLORD embedding generation, and hybrid FAISS/Elasticsearch search with BGE reranking — powering the agent's custom LangChain-style reasoning and tool-calling.
- Developed a minimal browser UI and toolchain for chat, QA/quiz, and PDF/DOCX/PPTX export.

Multilingual SLM | *HuggingFace, PyTorch* *Sept. 2025*

- Built a **3B-token multilingual corpus** (English, Tamil, Mizo) from scratch with rigorous cleaning, normalization, deduplication, and deterministic train/val/test splits.
- Trained a Unigram tokenizer (96k vocab) with temperature-smoothed sampling to preserve Mizo.
- Pretrained a 140M-parameter Gemma3 (dense) LM, attaining train perplexity **2.74** at 50k steps.

- Average validation perplexity reduced from **1865** to **112** (EN: 110, TA: 187, Mizo: 38).
- Fine-tuned with LoRA on two reasoning tasks — FAQ generation: **12.7** perplexity score, and context-aware synonym substitution: **1.73** - showing excellent transfer and controlled editing behavior.

Vision Transformers | *PyTorch* *Apr. 2025*

- Implemented Vision Transformer (ViT) and Differential ViT on CIFAR-10 from-scratch.
- Achieved **83.69%** test accuracy after benchmarking 11 ViT Configurations, and **84.38%** test accuracy after extending to differential attention (DiffViT). Produced DINO maps and hyperparameter tuning.

Fully Convolutional Networks for Semantic Segmentation | *PyTorch, VGG* *Apr. 2025*

- Built FCN-32, 16, 8 with VGG-16 backbone and bilinear upsampling on a 13-class dataset.
- Achieved **83.3%** test mIoU on fine-tuned backbone, lesser with frozen counterparts.
- Optimized data pipelines and batch scheduling to improve training throughput and GPU utilization.

OCR CNN-RNN | *Python, NumPy, PyTorch* *Oct. 2024*

- Developed a hybrid Deep Learning pipeline for OCR tasks using CNNs for feature extraction and RNNs with integrated attention mechanisms: achieved **99.1%** ANCC (character-level accuracy).

Multi-Layer Perceptron (MLP) from Scratch | *Python, NumPy, Matplotlib* *Sept. 2024*

- Implemented MLP from scratch, that supports: customizable hidden layers, activation functions, and optimization methods, to build intuition for gradient flow and model training dynamics.

BPSK Modulation System | *MATLAB* *April 2024*

- Built a Binary Phase Shift Keying (BPSK) system for audio transmission over an AWGN and Noiseless channel, using rectangular and raised cosine pulse shaping modulation for ISI analysis.

Y86-64 ISA Processor | *Verilog* *Feb. 2024*

- Built a pipelined implementation of a Y86-64 instruction-level processor with intermediate registers to optimize instruction throughput along with hazard detection and solutions to decode the same.
- Validated functional correctness through waveform-based testbenches and timing verification.

4-bit ALU Design | *Verilog, NGSpice, MAGIC* *Nov. 2023*

- Implemented a 4-bit ALU starting with Verilog modules, progressing to gate-level simulations in NGSpice, and finally transistor-level designs (NMOS/PMOS) and layout verification using MAGIC.

Additional Experience

Debate Club, IIIT-H - President *Oct. 2023 - Oct. 2025*

- Grew the club to **50+ active members**, with weekly debates and theory sessions.
- Built specialized teams for logistics, marketing, and design for efficient event management to handle logistics for the increased participation; tournament training for national competitions.

TEDx Talk: "Clarity in a Changing World" *March 2022*
IDEOU Foundations in Design Thinking with applied user research projects *August 2021*

Skills

- **Languages:** Python, C, C++, R, JavaScript, HTML, CSS, LaTeX
- **Software Engineering:** Git, Linux/Unix Development, FastAPI, REST API, Testing & Debugging
- **ML/AI:** PyTorch, HuggingFace, Transformers, XGBoost, Model Evaluation & Optimization
- **MLOps & Cloud:** Google Cloud Platform (BigQuery, Vertex AI, Pub/Sub, Cloud Build), Kubeflow, Docker, CI/CD Pipelines, Containerized Deployment
- **Tools:** MongoDB, Pandas, NumPy, Scikit-Learn, Docker, Jupyter
- **Soft Skills:** Public speaking, Technical writing, Team leadership, Cross-functional communication